Deploying Research:
On Building and Shipping an Anonymous Whistleblowing System

Tuesday, 10th March 2026

Taipei, Taiwan

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UNIVERSITY OF CAMBRIDGE

The free press fulfills an important function in a democracy.

Two workshops (2018) with real users revealed the first-contact problem.

FIRST-CONTACT PROBLEM

- Strong **metadata protection** from the very-first message.
- **Building trust** between the journalist and the source through **low-latency** communication.

"It is rare for it to be a high degree of emphasis placed on security by the source at that stage", so communication often "moves to another channel"

—P1 & P4

"[sources] sometimes become whistleblowers because they're getting increasingly angry about something"

—P9

CoverDrop is embedded in a news app.

“UNOBSERVABLE” COMMUNICATION

- **Encrypted** outer message with E2EE inner message.
- **Padded** to fixed size.
- **Sending pattern** independent of usage.

CoverDrop is embedded in a news app leveraging the large number of regular users to provide sources with strong plausible deniability.

NEWS ORGANISATION

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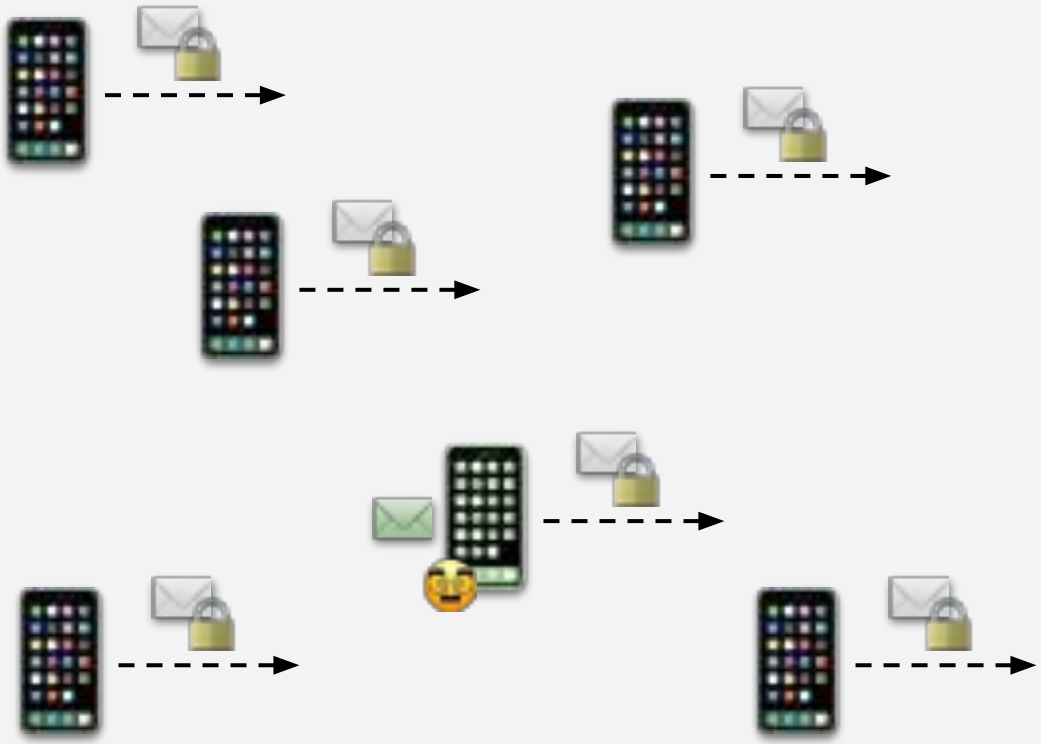

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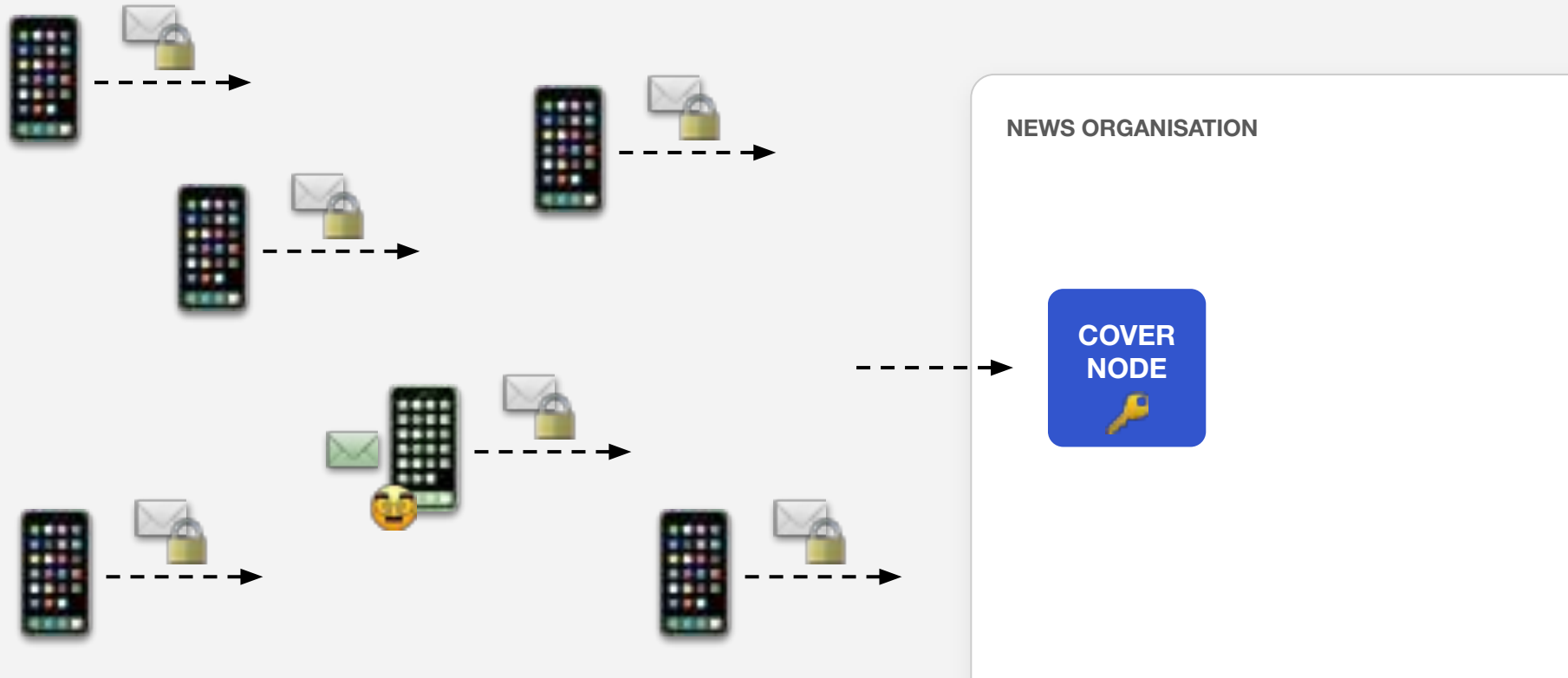

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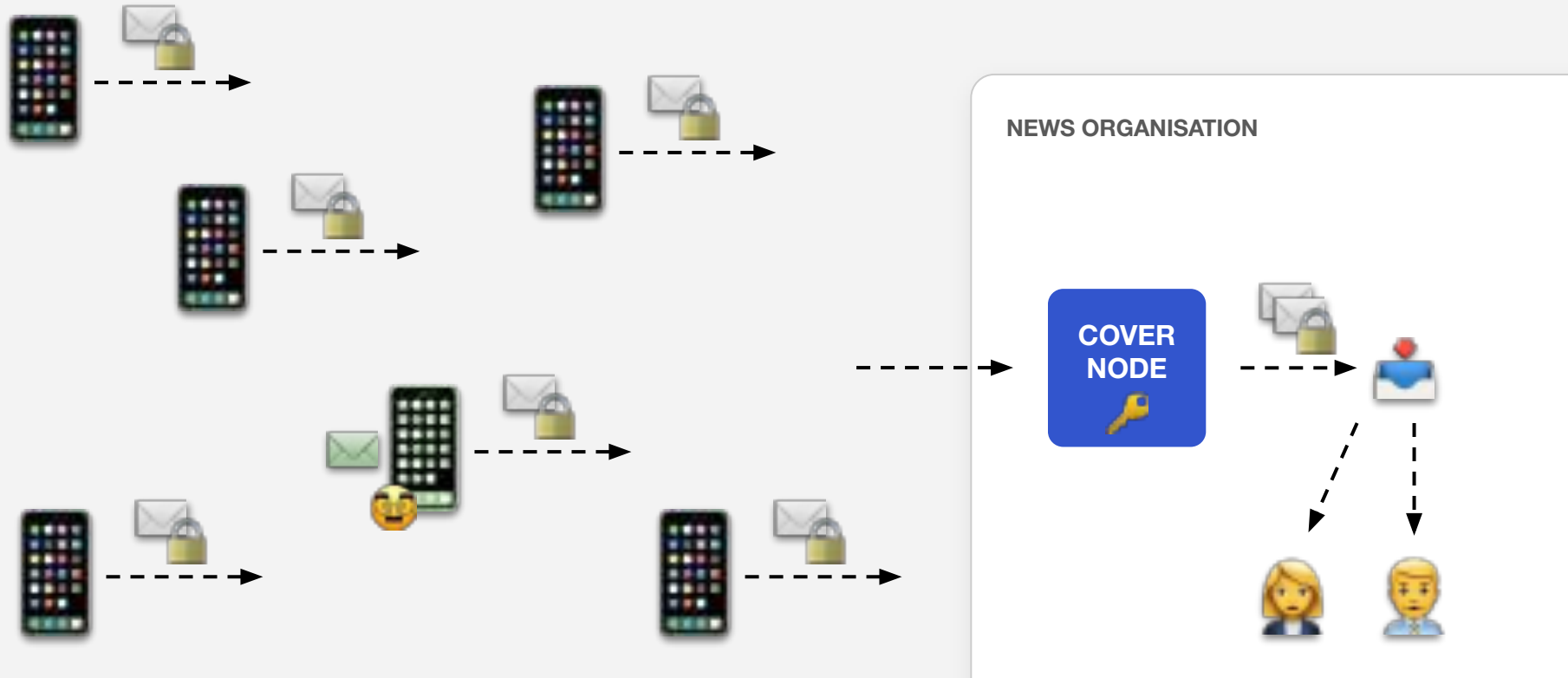

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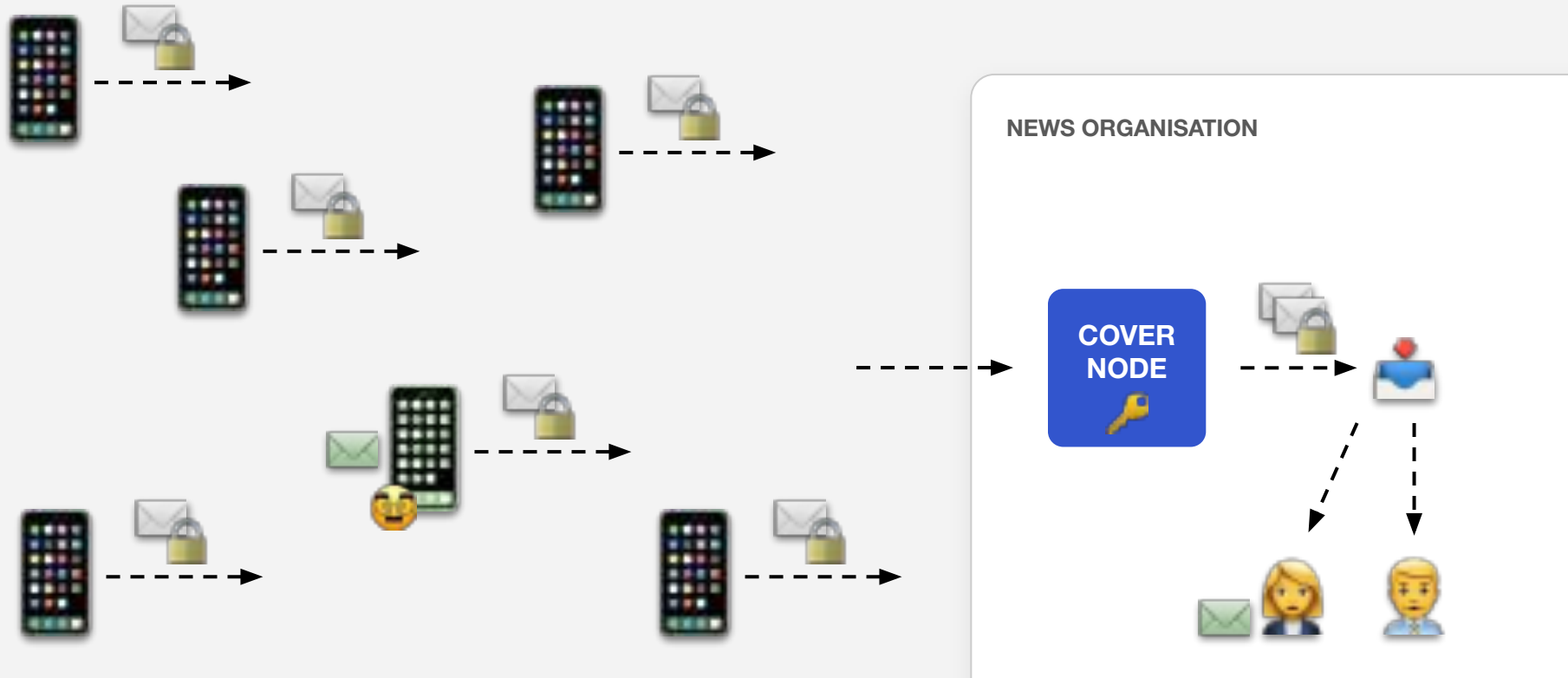

CoverDrop: Blowing the Whistle Through A News App

PETS '22

Mansoor Ahmed-Rengers

Diana A. Vasile

Alastair R. Beresford

Daniel Hugenroth

Ross Anderson

CoverDrop: Blowing the Whistle Through A News App

1 Introduction

Hey, this looks really interesting. Let’s build this! Shouldn’t take more than a few months…

—The Guardian

Hey, this looks really interesting. Let’s build this! Shouldn’t take more than a few months…

—The Guardian

Launched **June 2025** on Android and iOS

Expert input for the **updated white paper.**

Full audit sponsored by the OpenTech Fund.

Photo by https://commons.wikimedia.org/wiki/File:The_Guardian_Building_in_London.JPG (CC BY-SA)

CLICK-THROUGH

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Guardian sections
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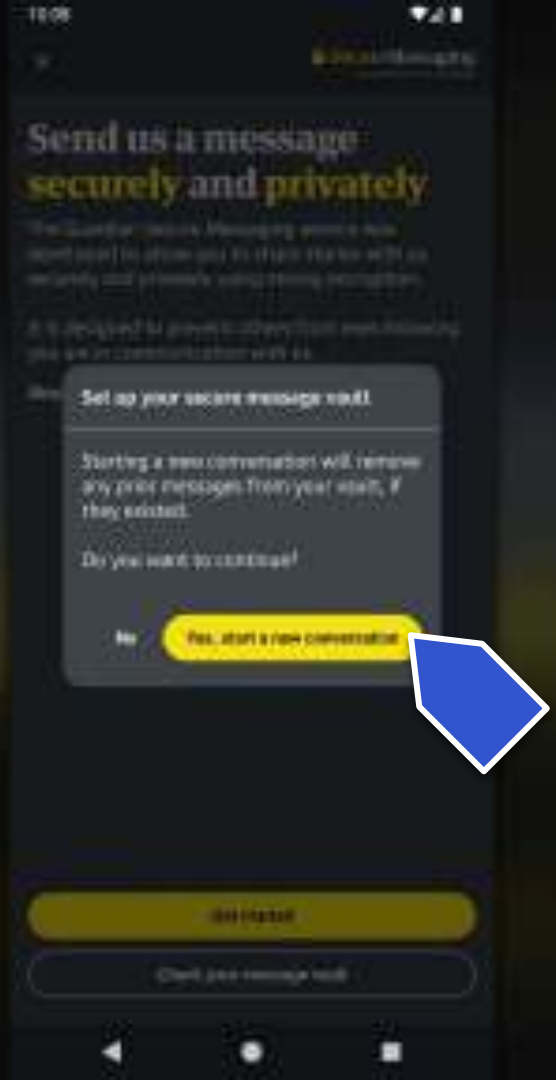
UK news
World news
Environment
Business
Technology
Science

Menu
Law
Opinion
Sport
Culture
Lifestyle
More...
Guardian services
Secure Messaging
Guardian Feast

SecureMessaging
Send us a message securely and privately
The Guardian Secure Messaging service was developed to allow you to share stories with us securely and privately using strong encryption.
It is designed to prevent others from even knowing you are in communication with us.
About Secure Messaging
Get started
Check your message vault


Send us a message
securely and privately
Set up your secure message vault
Starting a new conversation will remove any prior messages from your vault, if they existed.
Do you want to continue?
No
Yes, start a new conversation

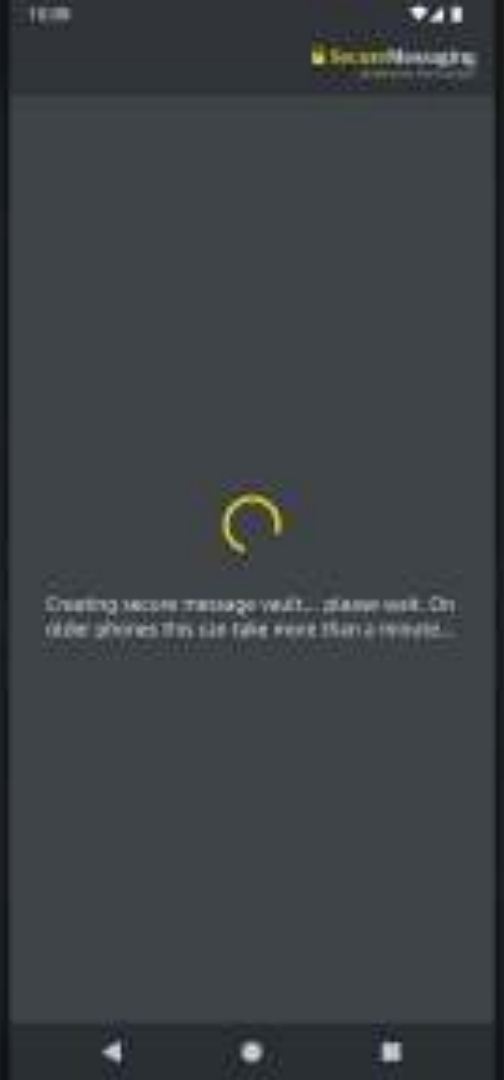
How this works
Send a message

How this works
Check back for a response
Our journalists check for new messages regularly. If they wish to take your story further they will respond here.
Continue

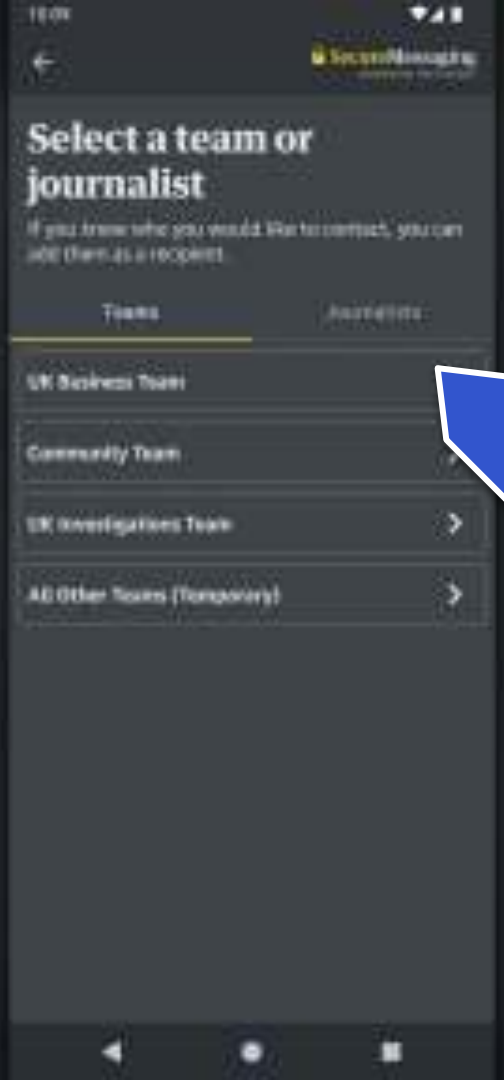
How this works
Memorise your passphrase

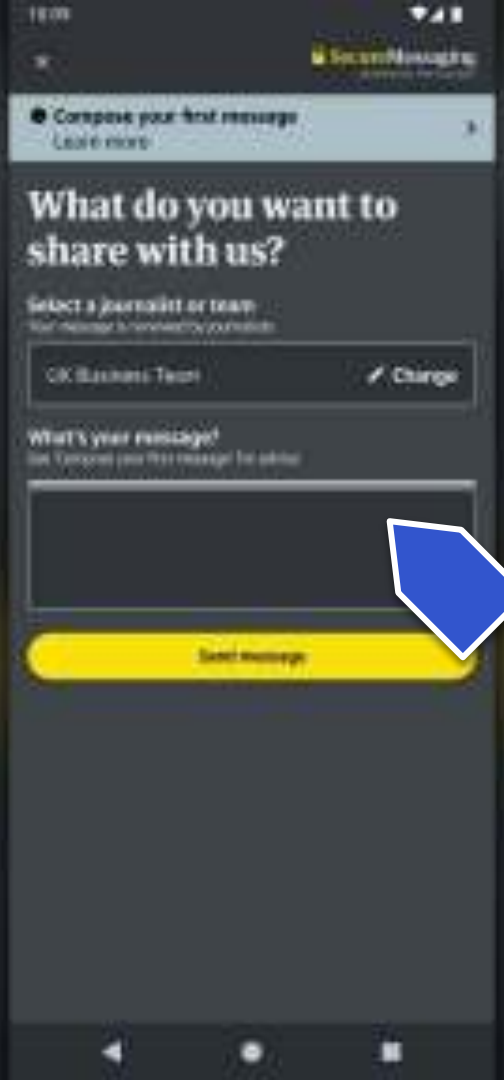
Keeping passphrases safe
Learn more
Remember passphrase
You will always need this passphrase to access your conversation. Make sure you memorize it or write it somewhere safe. Your passphrase must be entered in the correct order with the correct spelling.
harness
water
strawberry
Hide passphrase
I have remembered my passphrase
I already have a passphrase

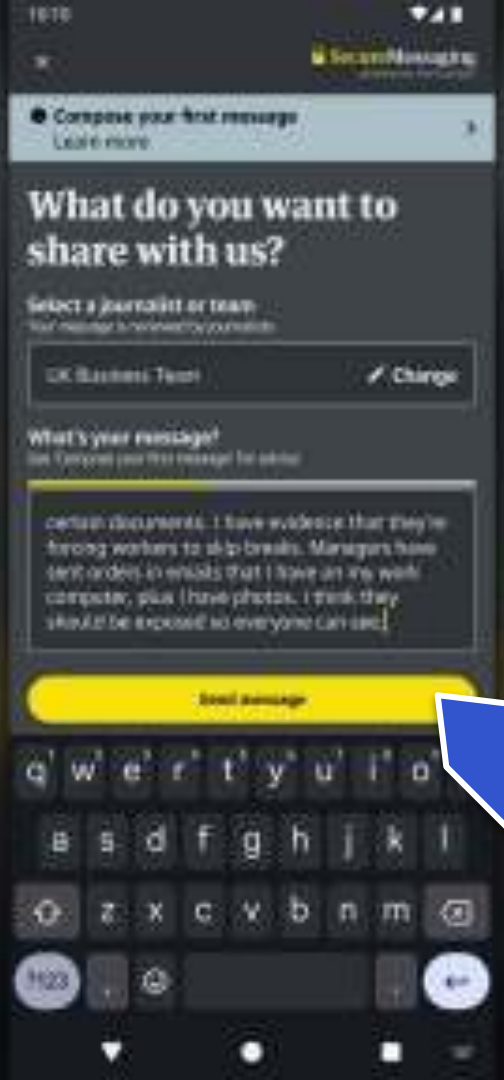
Enter passphrase
banana
water
Hide passphrase
Confirm passphrase


SecureMessaging
Creating secure message vault... please wait. On older phones this can take more than a minute...

Compose your first message
Learn more
What do you want to share with us?
Select a journalist or team
No recipient selected
Change
What's your message?

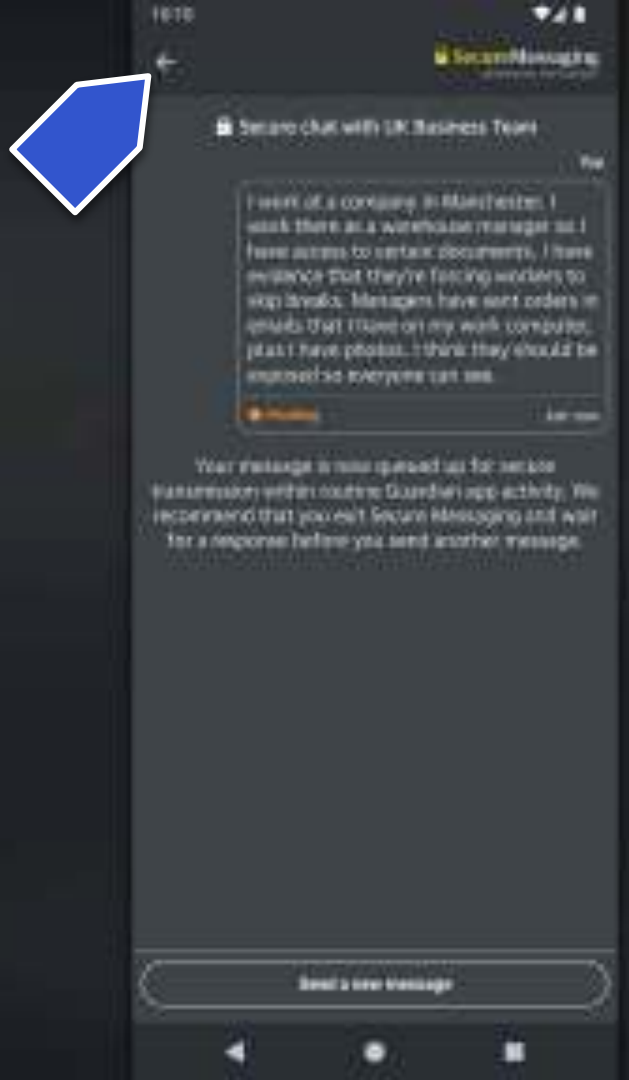

Select a team or journalist
If you know who you would like to contact, you can add them as a recipient
Teams
Journalists
UK Business Team
Community Team
UK Investigations Team
All Other Teams (Temporary)


Compose your first message
Learn more
What do you want to share with us?
Select a journalist or team
Change
What's your message?


Compose your first message
Learn more
What do you want to share with us?
Select a journalist or team
UK Business Team
Change
What's your message?
certain documents. I have evidence that they're forcing workers to skip breaks. Managers have sent orders in emails that I have on my work computer, plus I have photos. I think they should be exposed so everyone can see.
Send message

Your message will be received by a journalist soon
What happens next?
What to expect as a reply
Read more
Review conversation
Log out from Secure Messaging

UK Business Team


Secure chat with UK Business Team
I work at a company in Manchester. I work there as a warehouse manager so I have access to certain documents. I have evidence that they're forcing workers to skip breaks. Managers have sent orders in emails that I found on my work computer, plus I have photos. I think they should be exposed so everyone can see.
Your message is now queued up for secure transmission within routine Guardian app activity. We recommend that you exit Secure Messaging and wait for a response before you send another message.
Send a new message

Messaging with
UK Business Team
Delete message vault
About Secure Messaging
Leave message vault

ONE DAY LATER…

Send us a message
securely and privately
Get started
Check your message vault

10:19
SecureMessaging
Enter passphrase
If you previously set up a secure message vault, please enter your passphrase to unlock it.
Hide passphrase
I do not have a passphrase yet

Enter passphrase
Hide passphrase
Confirm Passphrase
I do not have a passphrase yet

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10:19
Unlocking secure vault... please wait. On older
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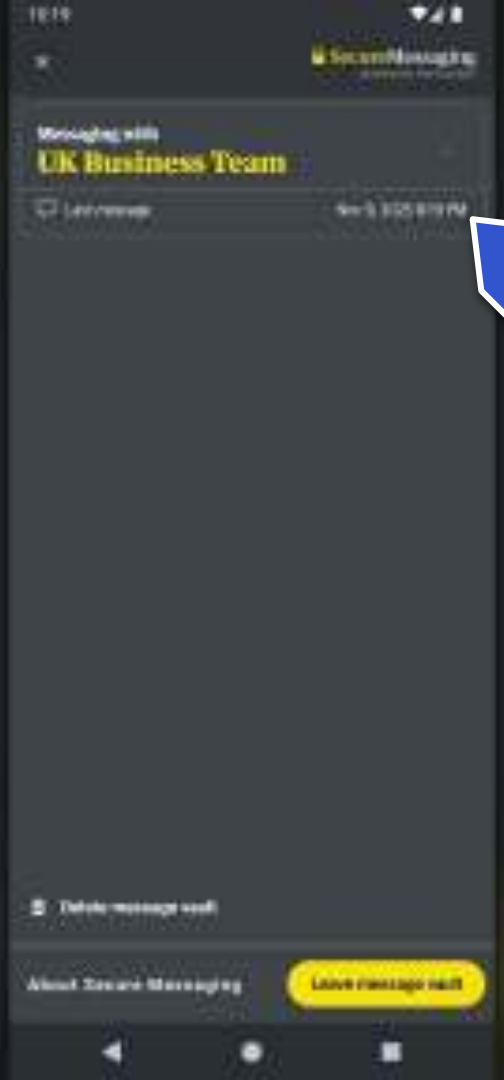

UK Business Team

10:19
SecureMessaging
Secure chat with UK Business Team
You
I work at a Company in Manchester. I work there as a warehouse manager so I have access to certain documents. I have evidence that they're forcing workers to skip breaks. Managers have sent orders in email on my work computer, plus I have photos. I think they should be exposed so everyone can see.
UK Business Team
Thank you for reaching out! My name is Joe Bloggs and I have worked on similar stories before. It is great that you have evidence. Can you tell me how long this has been going on?
2 hours ago
Send a new message

10:20

SecureMessaging

[redacted]

You

I work at a Company in Manchester. I work there as a warehouse manager so I have access to certain documents. I have evidence that they're forcing workers to skip breaks. Managers have sent orders in email on my work computer, plus I have photos. I think they should be exposed so everyone can see.

Apr 8, 2025 10:28 AM

[redacted]

Thank you for reaching out! My name is Joe Bloggs and I have worked on similar stories before. It is great that you have evidence. Can you tell me how long this has been going on?

2 hours ago

You

Almost six months by now. Can you let me know how I can send those photos to you? Is sending them via email a good idea?

[redacted] GMT 09:00

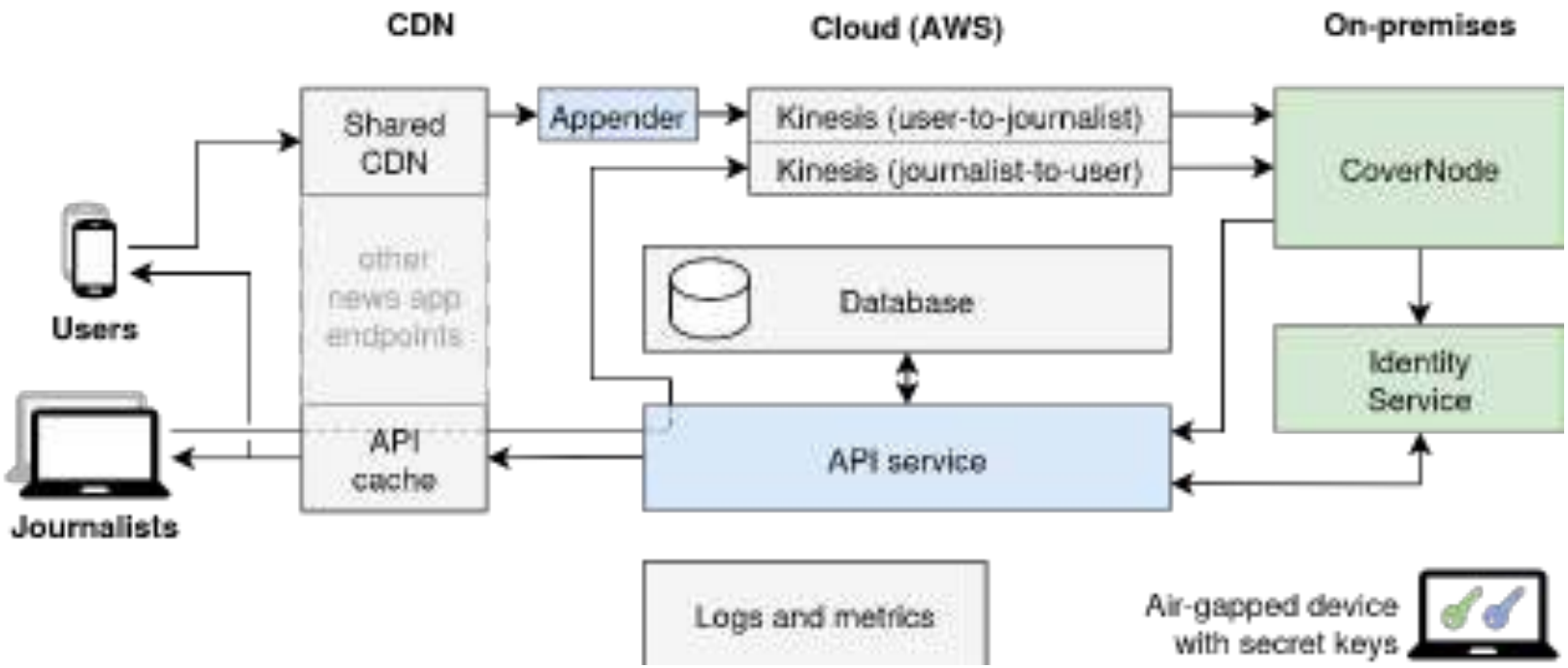
Your message is now queued up for secure transmission within mobile Guardian app activity. We recommend that you exit Secure Messaging and wait for a response before you send another message.

Send a new message

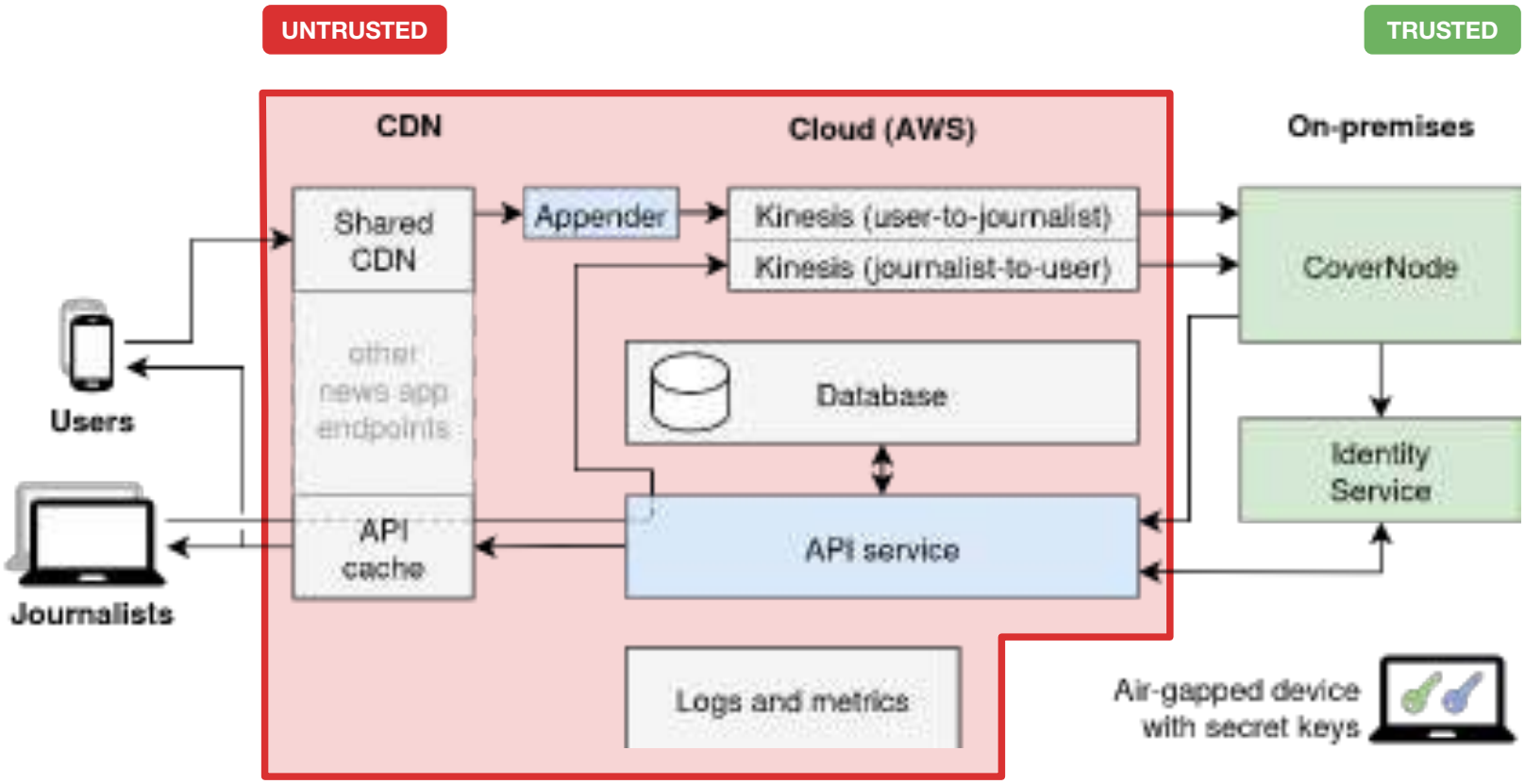
◀ ● ■

Sounds simple, doesn’t it?

A real-world system introduces real-world complexity.



Our threat model assumes minimal trust in external parties.



Managing complexity to build a reliable and secure system.

BORING CRYPTO

- Our protocol relies solely on **libsodium** primitives.
- Forward-secrecy through key rotation.
- **Minimize trusted scope.**

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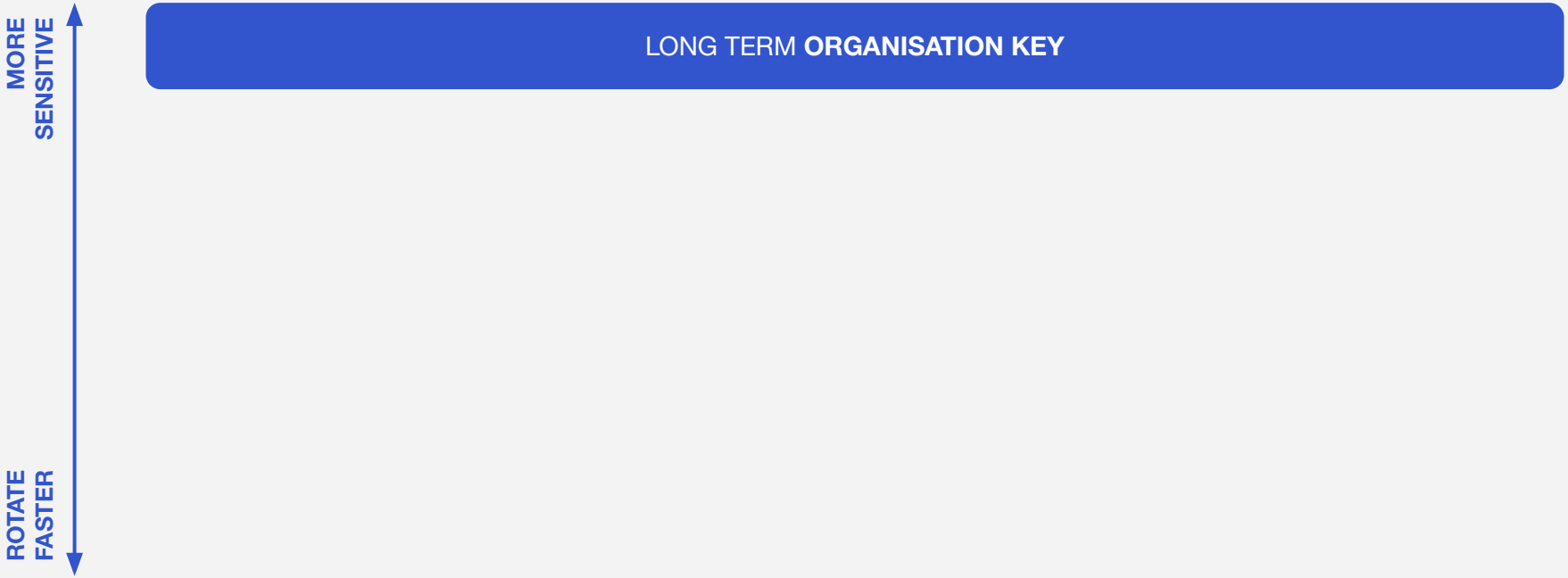
RUST SUCCESS STORY

- Leveraging types for mistake-resistant **API design.**
- Making **performance** a non-issue to afford simpler code.

See our RustConf talk for more info

Bootstrapping trust when there are no trusted third-party CAs.



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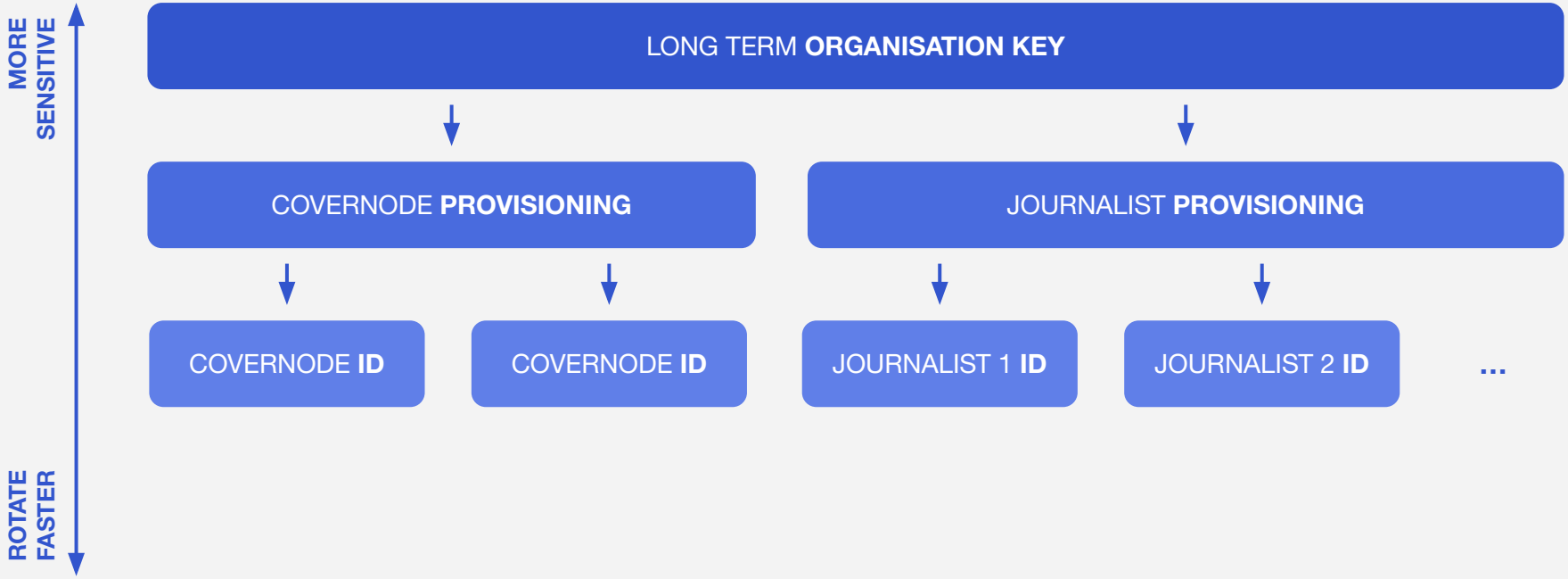

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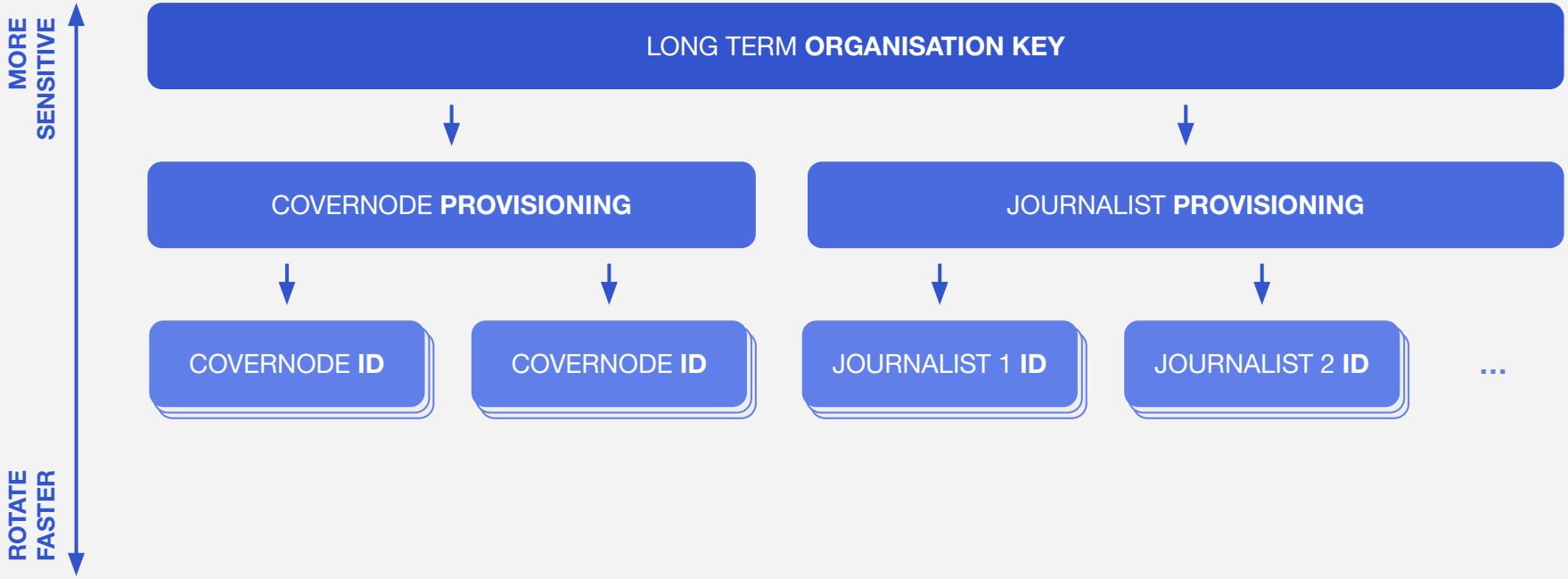

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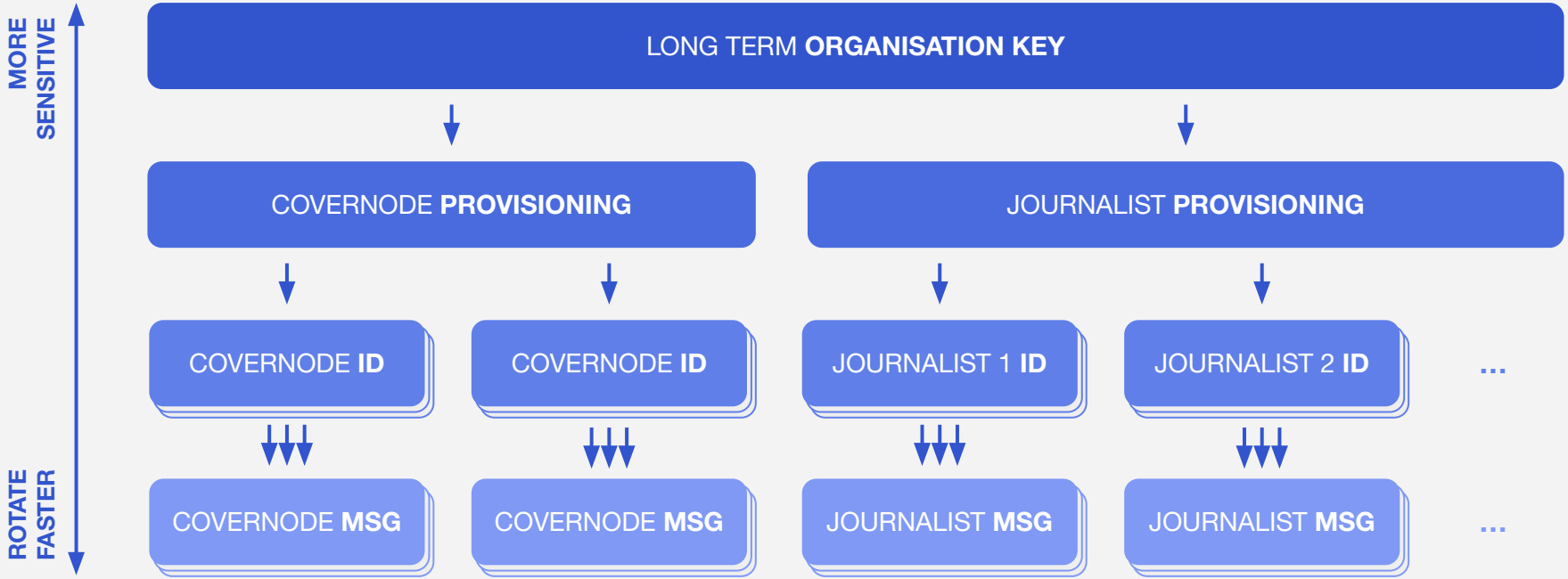

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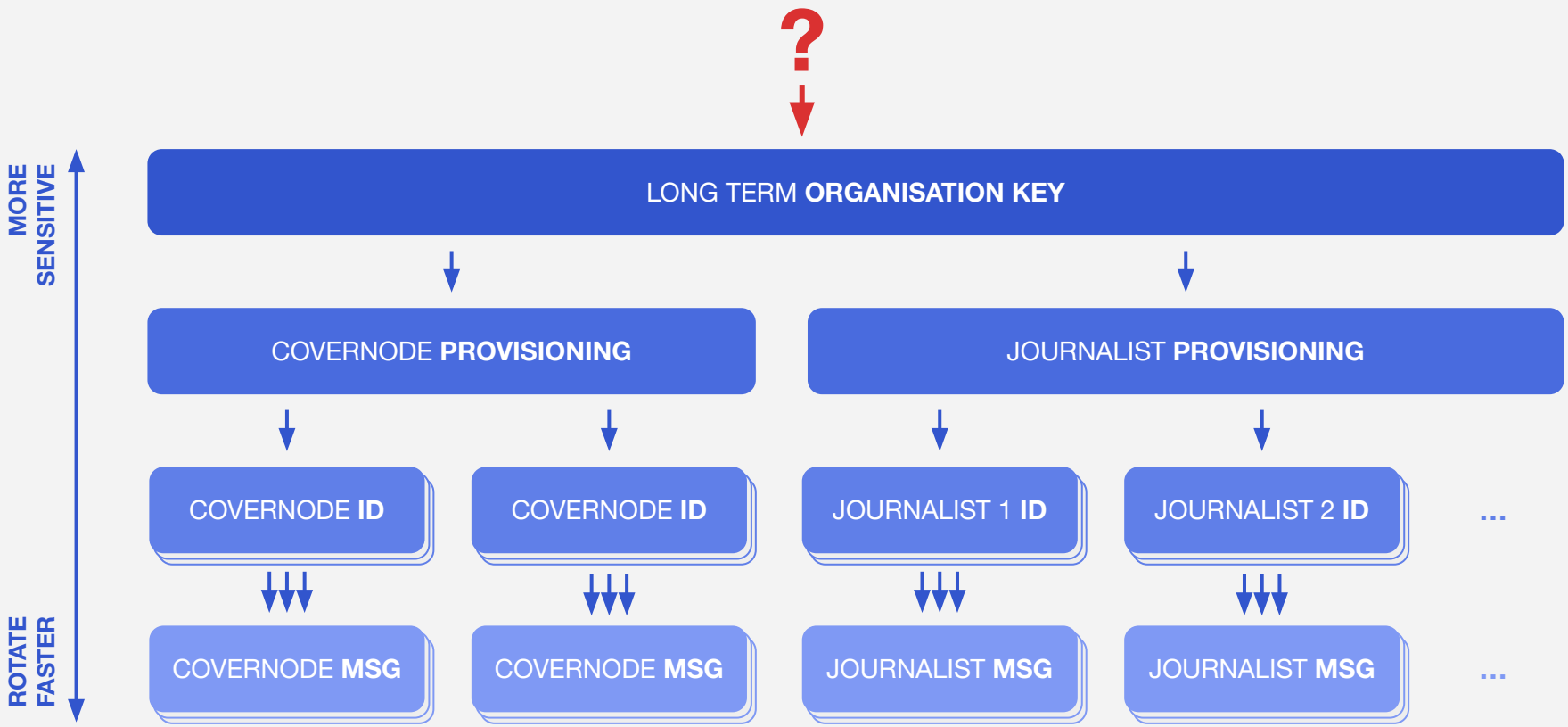

A printed out-of-band root of trust for the key hierarchy.

Photo by Grigorii Shcheglov on Unsplash

Words matter and good UX supports users in critical moments.

VERSION 1

Words matter and good UX supports users in critical moments.

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Words matter and good UX supports users in critical moments.

VERSION 1

VERSION 2

Turns out: in critical situations, users indeed read the manual.

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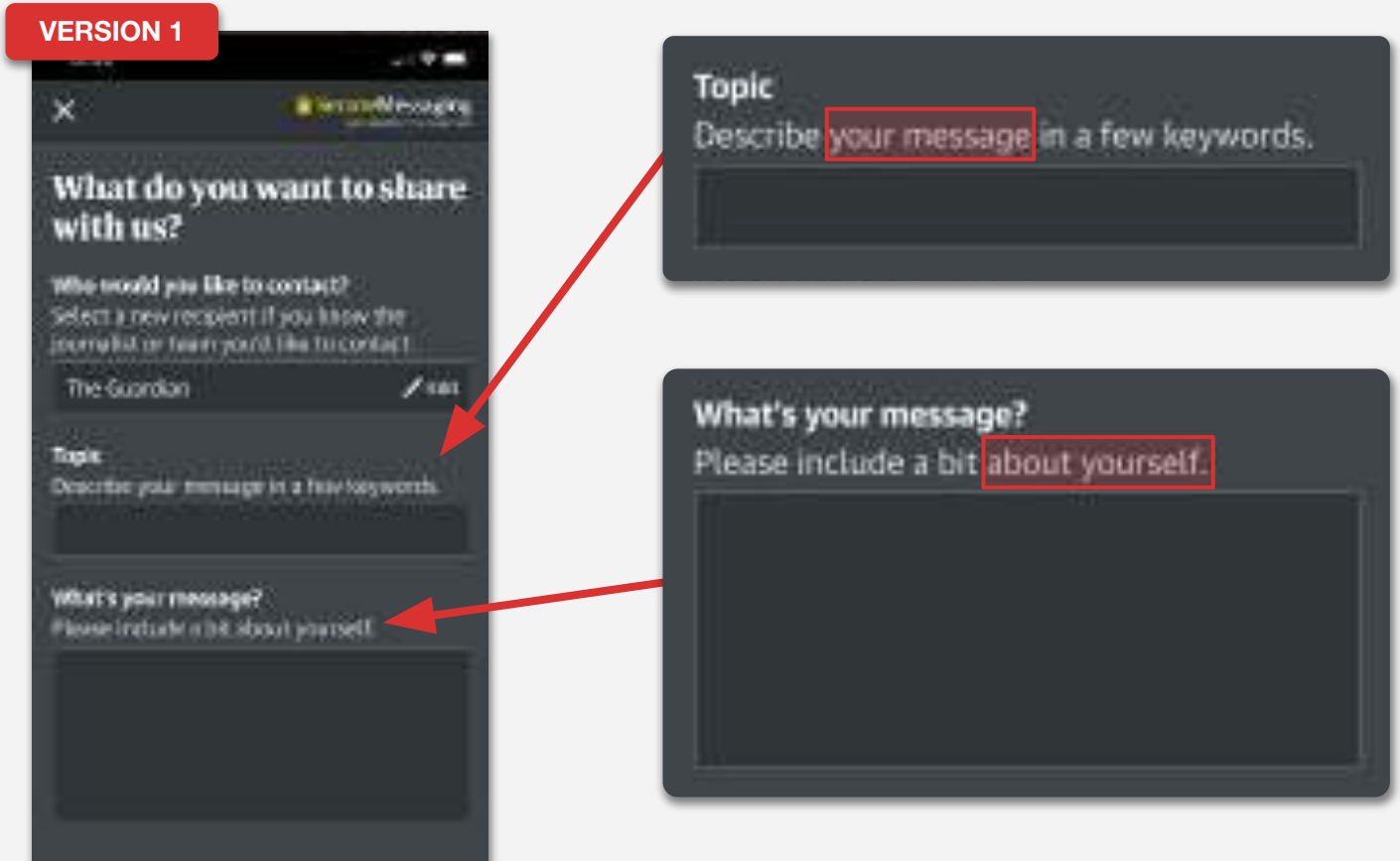

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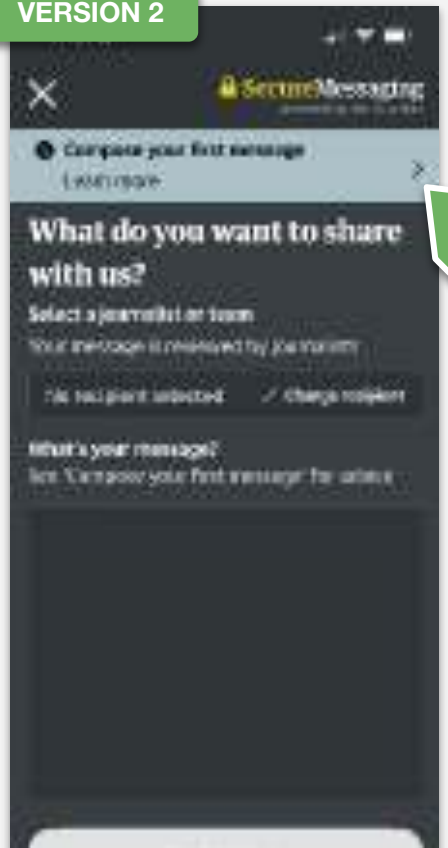
VERSION 2



Compose your first message

> 'We receive hundreds of tips a day. The messages that stand out are clear, detailed and fact-focused.'
>
> **Paul Lewis**
> Head of investigations

Messages are most credible when they are packed with facts, refer to evidence for those facts, and provide some context for how you know this.

Include what's safe

Facts

Specifics are critical – such as names, locations and dates.

> I work at government department 'Department name' in the procurement section.

Clearly state the nature of your tip

Be concise and focus on what you know, rather than what you think or suspect.

> I have discovered evidence of a senior official's involvement in corruption and bribery. 'Name', the head of our section, has received $150,000 in under-the-radar payments.

Evidence

Tell us how you know what you know, and how we might verify it.

> I have photographic evidence of transactions from the past two years and access to further documents that show there were a quid pro quo for awarding contracts to multinational company 'Company name'.

Our journalists appreciate that sharing information can be intimidating. You should only share what is best for your safety. However, vague messages are rarely actionable.

> 'You and your information will be treated with respect. Good sources are extremely important to us, and protecting their anonymity is a top priority.'
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> 'You and your information will be treated with respect. Good sources are extremely important to us, and protecting their anonymity is a top priority.'
>
> **Nick Hopkins**
> Head of news

“The quality of incoming messages is incomparable to what came before.”

—Deputy Investigations Editor

Making passphrase-based on-device encryption usable.

Making passphrase-based on-device encryption usable.

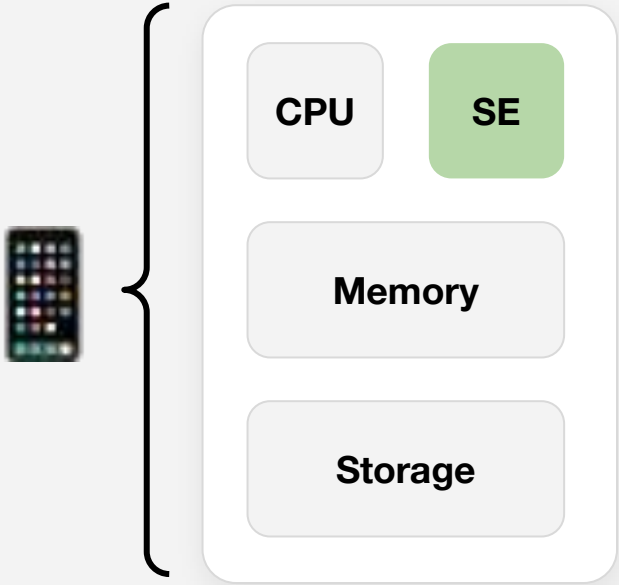
Wait! This is not sufficient min-entropy…

Making passphrase-based on-device encryption usable.

Wait! This is not sufficient min-entropy…

PETS '24

Secure Elements (SEs) are a great tool.



Secure Elements (SEs) are a great tool.

CPU

SE

Memory

Storage

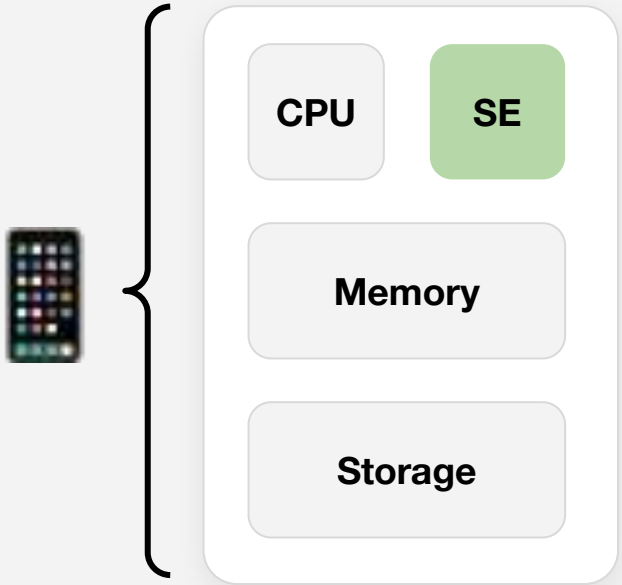
Protection against physical attacks

Authentication attempt counters

Internal keys and cryptographic operations

Enforcing time-outs

Secure Elements (SEs) are a great tool… or not?



Protection against physical attacks

Authentication attempt counters

Internal keys and cryptographic operations

Enforcing time-outs

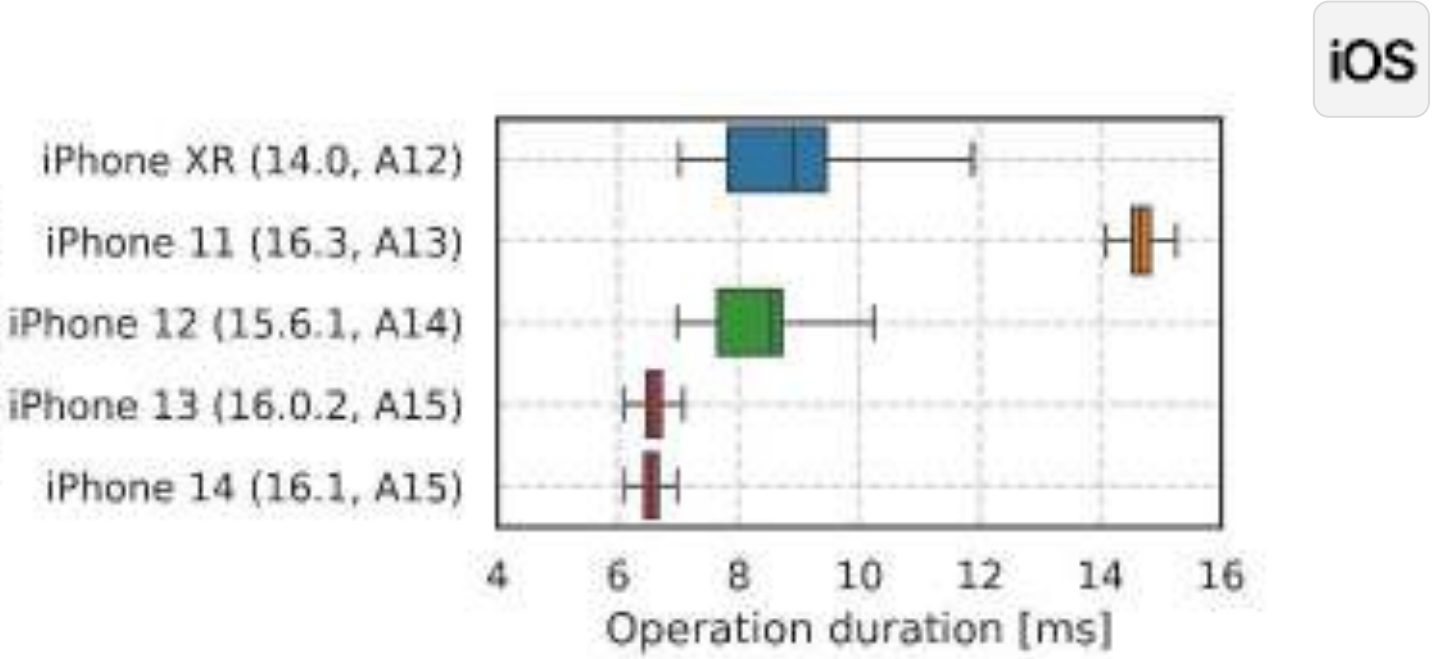
The limited SE APIs on smartphones.

- AES 128/256
- HMAC-SHA256
- 3DES
- RSA 2048
- ECDSA, ECDH P-256

- ECDSA, ECDH P-256

That's all...

Leveraging a bottleneck.



Iterative ECDH-powered key stretching.

procedure RainbowSloth.Derive(π, ψ, pw)

$l \leftarrow n \cdot (\frac{256}{8})$

$\omega_{pre,1} \parallel \cdots \parallel \omega_{pre,n} \leftarrow \text{PwHash}(\pi.salt, pw, l)$

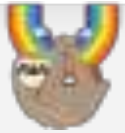
for i = 1 ... n **do**

$x \leftarrow \text{ReHashToP256}(\omega_{pre,i})$

$\omega_{post,i} \leftarrow \text{SE.Ecdh}(\psi, \pi.h, x)$

$k \leftarrow \text{Hkdf}(\omega_{post,1} \parallel \cdots \parallel \omega_{post,n})$

return k



Quo vadis: trustworthy digital artefacts, …

TRUSTING BINARY ARTEFACTS

- Establish trust in local and remote software.
- **Attestable Builds** make digital artefacts verifiable.

Light Squares .dev

ACM CCS '25

Quo vadis: trustworthy digital artefacts, improving cryptography foundations, and opportunities for collaboration.

TRUSTING BINARY ARTEFACTS

- Establish trust in local and remote software.
- **Attestable Builds** make digital artefacts verifiable.

CRYPTO NEXT STEPS

- Newsroom-internal **group communication** between journalists.
- **Formal verification** and revisiting in-depth audits.
- Post-quantum readiness.
- Didn't make the talk: private sending queue, SE-enabled ratcheting of plausibly-deniable ciphertexts, …

Thank you very much!
We have time for questions.

The original CoverDrop research team

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Ross Anderson

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www.coverdrop.org